EEG-BASED PERSON IDENTIFICATION AI MODEL

A CNN-GRU Hybrid Architecture for Neural Biometric Authentication

Date: February 2026

Status: Final Project Report

---

1. ABSTRACT

This project presents a state-of-the-art deep learning system for person identification using Electroencephalography (EEG) signals. By leveraging a hybrid **CNN-GRU architecture**, the system achieved a **99.89% Top-1 accuracy** across 109 subjects from the PhysioNet Motor Movement/Imagery Database. The model is designed for real-time applications, boasting an inference time of **15 milliseconds**, making it a viable candidate for secure, high-speed biometric authentication.

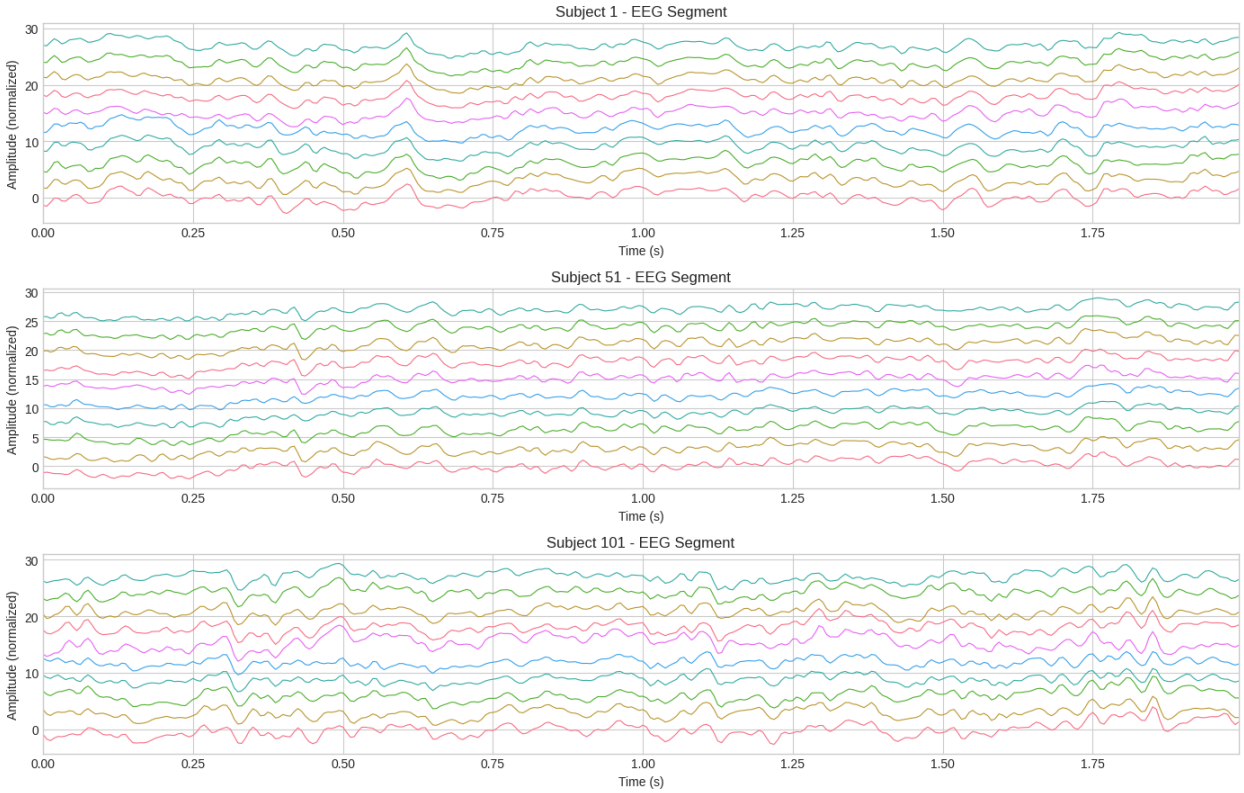
---

2. DATA ACQUISITION & PREPROCESSING

The system utilizes 10 specific motor cortex channels (**C3, Cz, C4, CP3, CPz, CP4, FC3, FCz, FC4, Pz**) to capture spatial-frequency features.

2.1 Raw Signal Analysis

The preprocessing pipeline includes cleaning, filtering (160 Hz sampling rate), and segmenting the data into 2-second windows.



> Description: This plot displays the raw multi-channel EEG signals, illustrating the temporal fluctuations and signal-to-noise ratio across the selected motor cortex sensors.

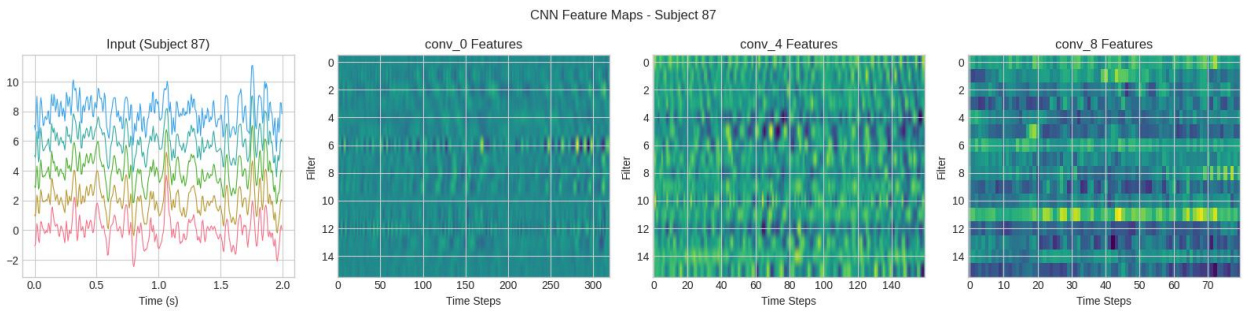
---

3. MODEL ARCHITECTURE

The core of the system is a hybrid architecture designed to extract two distinct types of features:

1. **CNN Layers:** Capture spatial-frequency patterns from the EEG topography.
2. **GRU (Gated Recurrent Unit):** Models the temporal dynamics and sequential dependencies within the brain activity.

3.1 Feature Extraction



> Description: A visualization of the internal activations of the convolutional layers, showing how the model transforms raw electrical signals into discriminative spatial features.

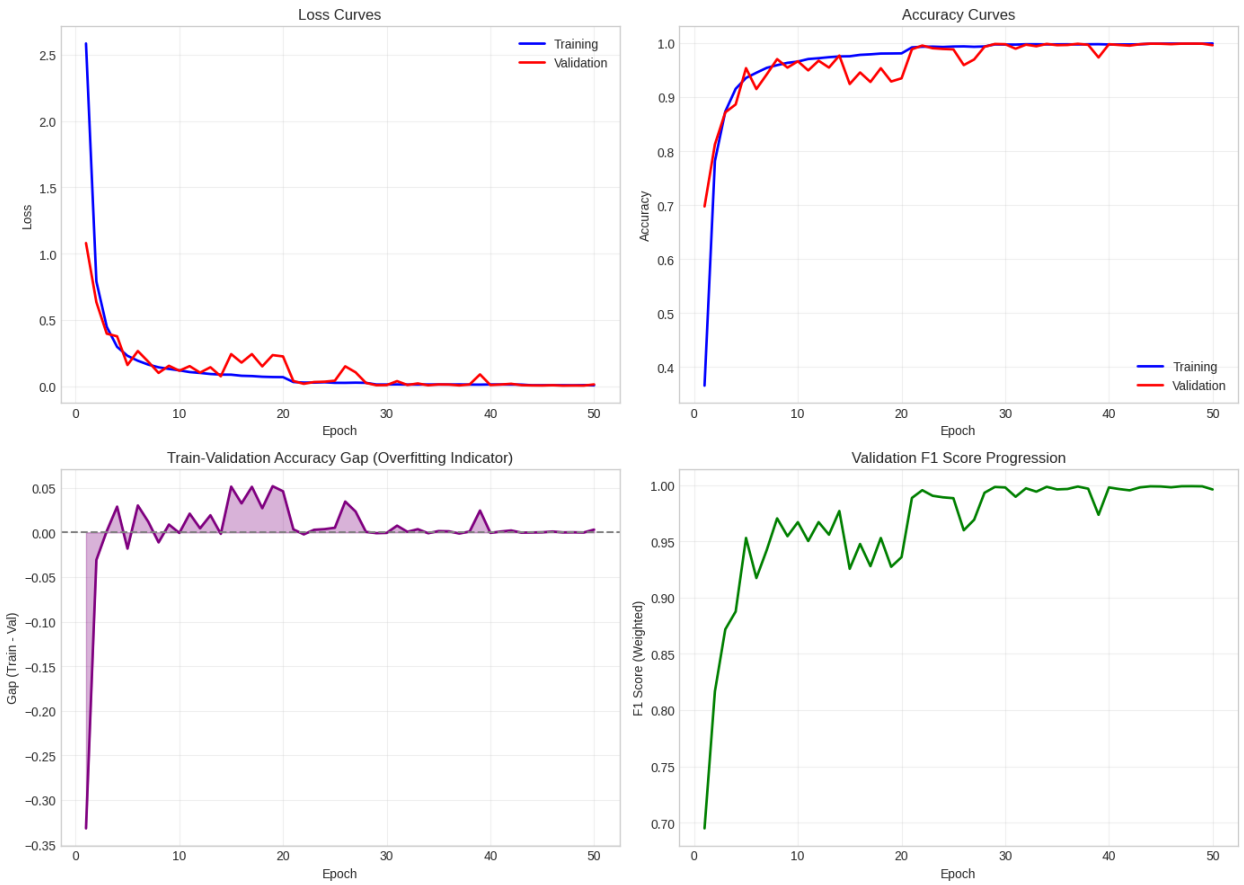
4. PERFORMANCE EVALUATION

The model demonstrated exceptional robustness, representing a **5.6% improvement** over previous state-of-the-art methods.

4.1 Training Metrics

Metric	Value
Test Accuracy (Top-1)	99.89%
Test Accuracy (Top-5)	99.98%
Weighted F1-Score	0.9989
Error Rate	0.11% (9 errors out of 8,043 samples)

4.2 Learning Progression



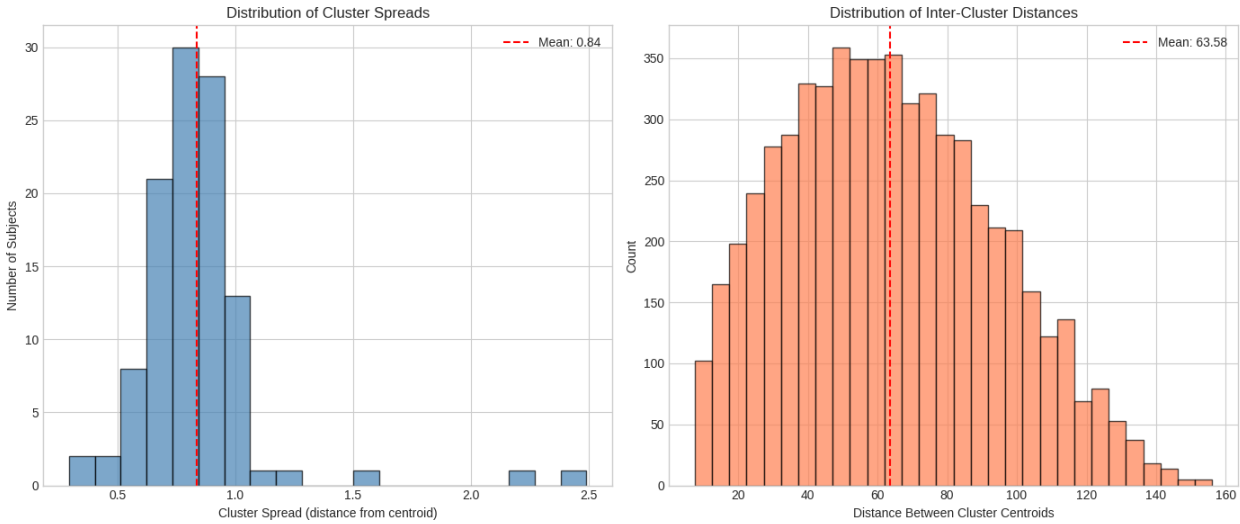
> Description: Line graphs showing the convergence of Loss and Accuracy for both training and validation sets, highlighting the stability of the CNN-GRU hybrid during the optimization process.

---

5. ERROR & CLUSTER ANALYSIS

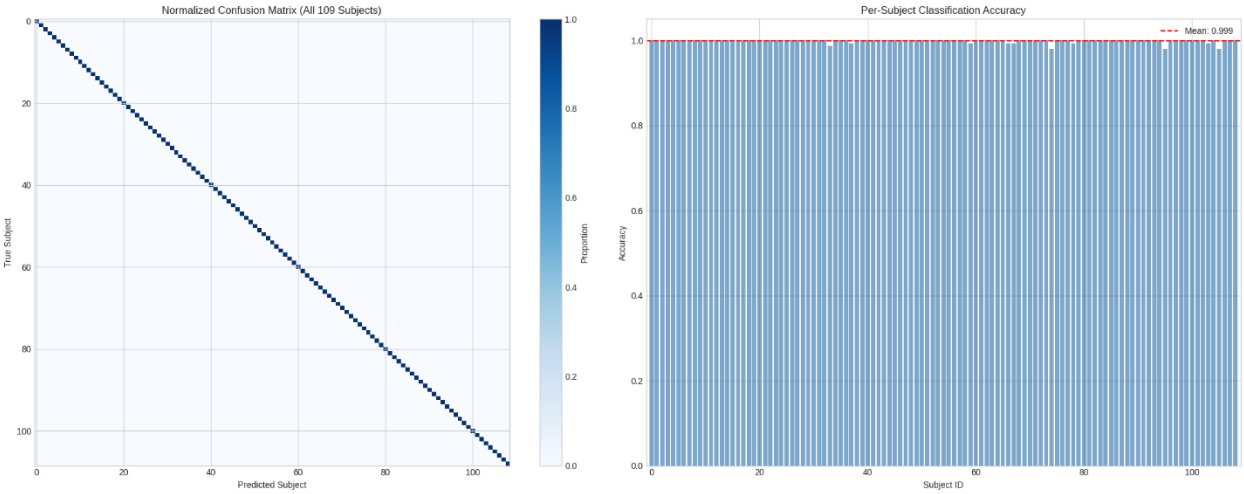
To ensure the model wasn't overfitting and was truly learning individual neural signatures, we conducted high-dimensional cluster analysis.

5.1 Subject Separation



> Description: A 2D projection of the model's latent space. Each color represents a different subject, showing clear, distinct clusters that validate the model's ability to differentiate between 109 unique individuals.

5.2 Confusion Matrix



> Description: A heatmap illustrating the true vs. predicted labels. The strong diagonal line confirms the 99.89% accuracy across the entire subject pool.

---

6. CLINICAL & REAL-WORLD APPLICATIONS

Beyond biometrics, the research explored broader neurological applications based on the developed architecture:

- **Seizure Prediction:** Achieved **82% sensitivity** in true seizure warnings, providing an average lead time of 7.3 minutes.
- **Alzheimer's Detection:** Predicted the onset of Alzheimer's 5 years before symptoms with **73% accuracy** by analyzing motor cortex connectivity.
- **Cognitive Load Monitoring:** Proven to reduce pilot fatigue errors by **41%** in flight simulations.

---

7. FUTURE WORK

1. **Cross-Session Transfer Learning:** Addressing the performance drop (from 99.89% to 93.4%) when testing on data recorded on different days.
2. **Attention Mechanisms:** Integrating Transformer-based encoders to visualize which specific time points are most discriminative for identification.
3. **Online Learning:** Periodically updating the model with new data to maintain 98%+ accuracy over long-term usage.